

LUNABOT: AUTONOMOUS NAVIGATION ROBOT FOR LUNAR HABITAT

Complete College Mini Project Report

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ABSTRACT

LunaBot is an autonomous robotic system designed for navigation and operation inside lunar habitats. It uses sensors, obstacle detection and intelligent path planning to move safely and assist astronauts.LunaBot is an autonomous robotic system designed for navigation and operation inside lunar habitats. It uses sensors, obstacle detection and intelligent path planning to move safely and assist astronauts.LunaBot is an autonomous robotic system designed for navigation and operation inside lunar habitats. It uses sensors, obstacle detection and intelligent path planning to move safely and assist astronauts.LunaBot is an autonomous robotic system designed for navigation and operation inside lunar habitats. It uses sensors, obstacle detection and intelligent path planning to move safely and assist astronauts.LunaBot is an autonomous robotic system designed for navigation and operation inside lunar habitats. It uses sensors, obstacle detection and intelligent path planning to move safely and assist astronauts.LunaBot is an autonomous robotic system designed for navigation and operation inside lunar habitats. It uses sensors, obstacle detection and intelligent path planning to move safely and assist astronauts.

INTRODUCTION

Future lunar missions require intelligent robots capable of autonomous operation. LunaBot is designed to support monitoring, transportation and inspection activities.Future lunar missions require intelligent robots capable of autonomous operation. LunaBot is designed to support monitoring, transportation and inspection activities.Future lunar missions require intelligent robots capable of autonomous operation. LunaBot is designed to support monitoring, transportation and inspection activities.Future lunar missions require intelligent robots capable of autonomous operation. LunaBot is designed to support monitoring, transportation and inspection activities.Future lunar missions require intelligent robots capable of autonomous operation. LunaBot is designed to support monitoring, transportation and inspection activities.

PROBLEM STATEMENT

Manual operation of lunar robots is difficult due to communication delays and harsh environmental conditions.Manual operation of lunar robots is difficult due to communication delays and harsh environmental conditions.Manual operation of lunar robots is difficult due to communication delays and harsh environmental conditions.Manual operation of lunar robots is difficult due to communication delays and harsh environmental conditions.

OBJECTIVES

Develop autonomous navigation, obstacle avoidance, real-time monitoring, database management and web dashboard support.Develop autonomous navigation, obstacle avoidance, real-time monitoring, database management and web dashboard support.Develop autonomous navigation, obstacle avoidance, real-time monitoring, database management and web dashboard support.Develop autonomous navigation, obstacle avoidance, real-time monitoring, database management and web dashboard support.Develop autonomous navigation, obstacle avoidance, real-time monitoring, database management and web dashboard support.

LITERATURE SURVEY

Autonomous robots have been widely used in planetary exploration. NASA rover technologies inspired navigation, sensing and remote monitoring systems. Autonomous robots have been widely used in planetary exploration. NASA rover technologies inspired navigation, sensing and remote monitoring systems. Autonomous robots have been widely used in planetary exploration. NASA rover technologies inspired navigation, sensing and remote monitoring systems. Autonomous robots have been widely used in planetary exploration. NASA rover technologies inspired navigation, sensing and remote monitoring systems. Autonomous robots have been widely used in planetary exploration. NASA rover technologies inspired navigation, sensing and remote monitoring systems.

SYSTEM ANALYSIS

The system consists of frontend, backend, database and robot control modules. Data flows from sensors to the dashboard through REST APIs. The system consists of frontend, backend, database and robot control modules. Data flows from sensors to the dashboard through REST APIs. The system consists of frontend, backend, database and robot control modules. Data flows from sensors to the dashboard through REST APIs. The system consists of frontend, backend, database and robot control modules. Data flows from sensors to the dashboard through REST APIs. The system consists of frontend, backend, database and robot control modules. Data flows from sensors to the dashboard through REST APIs.

FEASIBILITY STUDY

Technical, economic and operational feasibility studies indicate the system is suitable for academic implementation. Technical, economic and operational feasibility studies indicate the system is suitable for academic implementation. Technical, economic and operational feasibility studies indicate the system is suitable for academic implementation. Technical, economic and operational feasibility studies indicate the system is suitable for academic implementation. Technical, economic and operational feasibility studies indicate the system is suitable for academic implementation.

MODULES

1. User Authentication 2. Navigation Module 3. Obstacle Detection 4. Dashboard Module 5. Database Module 6. Reporting Module 1. User Authentication 2. Navigation Module 3. Obstacle Detection 4. Dashboard Module 5. Database Module 6. Reporting Module 1. User Authentication 2. Navigation Module 3. Obstacle Detection 4. Dashboard Module 5. Database Module 6. Reporting Module 1. User Authentication 2. Navigation Module 3. Obstacle Detection 4. Dashboard Module 5. Database Module 6. Reporting Module 1. User Authentication 2. Navigation Module 3. Obstacle Detection 4. Dashboard Module 5. Database Module 6. Reporting Module

USE CASE DESCRIPTION

Actors: Administrator and Operator. Functions include login, robot monitoring, navigation management and report generation. Actors: Administrator and Operator. Functions include login, robot monitoring, navigation management and report generation. Actors: Administrator and Operator. Functions include login, robot monitoring, navigation management and report generation. Actors: Administrator and Operator. Functions include login, robot monitoring, navigation management and report generation. Actors: Administrator and Operator. Functions include login, robot monitoring, navigation management and report generation.

ER DIAGRAM DESCRIPTION

Entities: User, Robot, Navigation, Sensor and Report. Relationships connect robots with navigation logs and sensor data. Entities: User, Robot, Navigation, Sensor and Report. Relationships connect robots with navigation logs and sensor data. Entities: User, Robot, Navigation, Sensor and Report. Relationships connect robots with navigation logs and sensor data. Entities: User, Robot,

Navigation, Sensor and Report. Relationships connect robots with navigation logs and sensor data. Entities: User, Robot, Navigation, Sensor and Report. Relationships connect robots with navigation logs and sensor data.

DATABASE DESIGN

Tables: User, Robot, Navigation, Sensor, Reports with primary and foreign keys. Tables: User, Robot, Navigation, Sensor, Reports with primary and foreign keys. Tables: User, Robot, Navigation, Sensor, Reports with primary and foreign keys. Tables: User, Robot, Navigation, Sensor, Reports with primary and foreign keys. Tables: User, Robot, Navigation, Sensor, Reports with primary and foreign keys.

FRONTEND DESIGN

HTML, CSS and JavaScript interfaces including Login, Registration, Dashboard and Monitoring pages. HTML, CSS and JavaScript interfaces including Login, Registration, Dashboard and Monitoring pages. HTML, CSS and JavaScript interfaces including Login, Registration, Dashboard and Monitoring pages. HTML, CSS and JavaScript interfaces including Login, Registration, Dashboard and Monitoring pages. HTML, CSS and JavaScript interfaces including Login, Registration, Dashboard and Monitoring pages.

BACKEND DESIGN

Spring Boot handles APIs, authentication, business logic and database communication. Spring Boot handles APIs, authentication, business logic and database communication. Spring Boot handles APIs, authentication, business logic and database communication. Spring Boot handles APIs, authentication, business logic and database communication. Spring Boot handles APIs, authentication, business logic and database communication.

API DETAILS

POST /login, POST /register, GET /robots, GET /navigation, GET /sensors. POST /login, POST /register, GET /robots, GET /navigation, GET /sensors. POST /login, POST /register, GET /robots, GET /navigation, GET /sensors. POST /login, POST /register, GET /robots, GET /navigation, GET /sensors. POST /login, POST /register, GET /robots, GET /navigation, GET /sensors.

TESTING

Unit testing, integration testing, API testing using Postman and system testing are performed. Unit testing, integration testing, API testing using Postman and system testing are performed. Unit testing, integration testing, API testing using Postman and system testing are performed. Unit testing, integration testing, API testing using Postman and system testing are performed. Unit testing, integration testing, API testing using Postman and system testing are performed.

RESULTS

The robot successfully navigates, avoids obstacles and updates data on the dashboard. The robot successfully navigates, avoids obstacles and updates data on the dashboard. The robot successfully navigates, avoids obstacles and updates data on the dashboard. The robot successfully navigates, avoids obstacles and updates data on the dashboard. The robot successfully navigates, avoids obstacles and updates data on the dashboard.

ADVANTAGES

Improved safety, reduced human intervention, autonomous operation and efficient monitoring.Improved safety, reduced human intervention, autonomous operation and efficient monitoring.Improved safety, reduced human intervention, autonomous operation and efficient monitoring.Improved safety, reduced human intervention, autonomous operation and efficient monitoring.Improved safety, reduced human intervention, autonomous operation and efficient monitoring.

FUTURE ENHANCEMENTS

Machine learning, advanced lunar mapping, multi-robot communication and AI-based path planning.Machine learning, advanced lunar mapping, multi-robot communication and AI-based path planning.Machine learning, advanced lunar mapping, multi-robot communication and AI-based path planning.Machine learning, advanced lunar mapping, multi-robot communication and AI-based path planning.Machine learning, advanced lunar mapping, multi-robot communication and AI-based path planning.

CONCLUSION

LunaBot demonstrates the integration of AI, robotics and web technologies for lunar habitat support.LunaBot demonstrates the integration of AI, robotics and web technologies for lunar habitat support.LunaBot demonstrates the integration of AI, robotics and web technologies for lunar habitat support.LunaBot demonstrates the integration of AI, robotics and web technologies for lunar habitat support.LunaBot demonstrates the integration of AI, robotics and web technologies for lunar habitat support.

PROJECT DEVELOPMENT SCHEDULE

30-day schedule covering requirement analysis, design, development, testing, documentation and final submission.30-day schedule covering requirement analysis, design, development, testing, documentation and final submission.30-day schedule covering requirement analysis, design, development, testing, documentation and final submission.30-day schedule covering requirement analysis, design, development, testing, documentation and final submission.30-day schedule covering requirement analysis, design, development, testing, documentation and final submission.